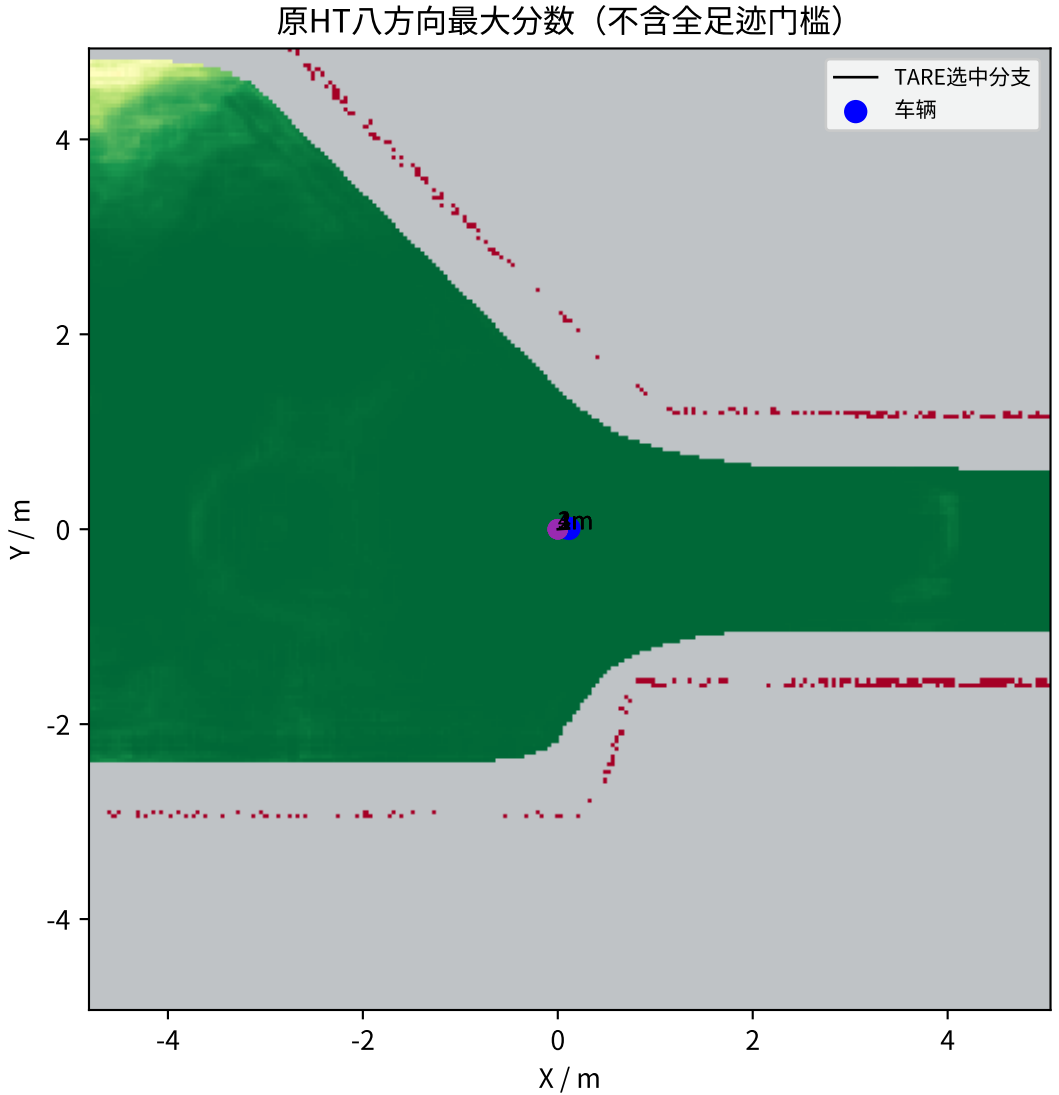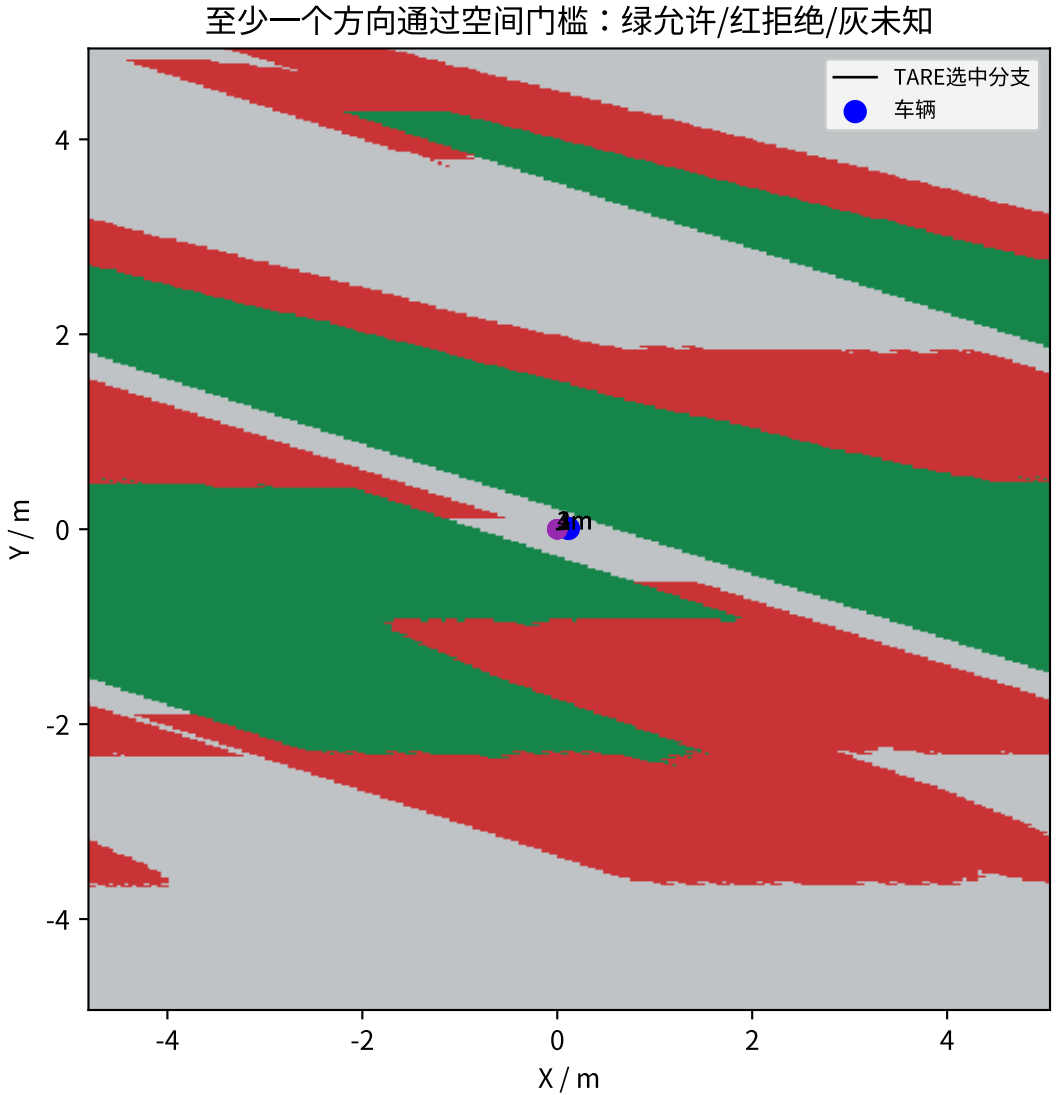记录快照t=586.8s：固定1/2/3/4m局部终点检查；不是搜索可达性证明